

CS560 - UI/UX Design & Implementation

7-1 Short Paper: Comparing Usability Testing Methods and Metrics

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Comparing Usability Testing Methods for an Online Banking Mobile App

Usability testing helps organizations understand how users interact with digital products and identify areas that need improvement. Choosing the right testing method is important because different methods provide diverse types of feedback and insights. For this paper, I selected an online banking mobile application focused on transferring money between accounts. This task is one of the most key features in a banking app because users expect it to be quick, accurate, and easy to complete. I will compare moderated usability testing and unmoderated remote usability testing, discuss their strengths and limitations, and explain which usability metrics would be most effective for evaluating this workflow.

Interface and User Task

The selected interface is a mobile banking application that allows customers to manage their accounts, pay bills, and transfer funds. The specific task being evaluated is transferring money from a checking account to a savings account. To complete this workflow, users must log in, navigate to the transfer section, select the accounts involved, enter the transfer amount, review the information, and confirm the transaction. This workflow is critical because errors can lead to financial consequences, and users expect the process to be efficient and trustworthy. Evaluating the usability of this task can help identify navigation issues, confusing instructions, or design elements that may prevent successful completion.

Moderated Usability Testing

Moderated usability testing involves a facilitator observing participants while they complete specific tasks. During the session, the moderator may ask questions, encourage participants to

think aloud, and gather feedback about the user's experience. For this workflow, participants would be asked to transfer money between accounts while the moderator observes their actions. The moderator could ask questions about what users expect to happen, whether any screens are confusing, and why they make certain decisions during the process. This method generates both qualitative and quantitative data. Observations may include user comments, frustrations, navigation patterns, and points where users hesitate. Quantitative data may include task completion rates, time on task, and the number of errors made during the process. One major strength of moderate testing is the ability to gain detailed insights into user behavior. If participants encounter problems, the moderator can ask follow-up questions to understand the reasons behind the issue. However, moderated testing can be time-consuming and expensive because each session requires a facilitator and scheduled participant sessions (Rubin & Chisnell, 2008).

Unmoderated Remote Usability Testing

Unmoderated remote usability testing allows participants to complete tasks independently without a facilitator present. Participants receive instructions and perform tasks using specialized testing software that records their interactions. For the banking app workflow, users would receive a scenario instructing them to transfer funds between accounts. The testing platform would record screen interactions, clicks, completion times, and survey responses after the task is completed. This method typically generates larger amounts of quantitative data because more participants can be tested in a shorter period. Researchers can analyze success rates, task completion times, click paths, and user satisfaction ratings.

A major advantage of unmoderated testing is its cost-effectiveness and scalability. Organizations can collect feedback from users across various locations without needing a facilitator. However, the method provides less context when users experience problems because there is no moderator available to ask questions or clarify observations (Barnum, 2020).

Comparison of the Methods

Both methods can evaluate the money transfer workflow effectively, but they serve different purposes. Moderated usability testing is best when the organization wants detailed information about why users experience difficulties. This method is especially useful during early design stages when designers need to identify usability problems and understand user behavior in depth. Unmoderated remote testing is more appropriate when the organization wants feedback from a larger group of users and needs results quickly. This method works well during later stages of development when designers want to validate improvements or measure overall usability performance. The main strength of moderated testing is the richness of the feedback collected. The main limitation is the higher cost and time investment. In contrast, unmoderated testing allows organizations to gather data from many participants efficiently, but it provides less detailed explanations for usability issues. In a professional setting, I would choose moderated usability testing when investigating complex usability problems or evaluating an early prototype. I would choose unmoderated remote testing when validating a finished design or collecting data from a large user population.

Usability Metrics and Analytics

For moderated usability testing, I would focus on metrics such as task completion rate, time on task, and the number of user errors made during the money transfer process. These measurements help determine whether users can successfully complete the task and how efficiently they can do so. In addition to quantitative data, I would pay close attention to participant comments, questions, and visible signs of frustration during the session. Because a moderator is present, it is possible to ask follow-up questions and gain a better understanding of why users experience difficulties. This combination of performance data and direct user feedback provides valuable insight into usability issues that may not be obvious from numbers alone. For unmoderated remote usability testing, I would focus on task completion rates, time on tasks, click paths, misclick rates, and post-task satisfaction ratings. Since participants complete the test independently, these metrics provide objective data about how users navigate the application and whether they can complete the transfer successfully. Click path analysis can reveal whether users are following the intended navigation flow, while misclick rates may identify confusing buttons, labels, or screen layouts. Post-task satisfaction ratings help measure users' overall impressions of the experience and whether they find the process easy to use. Together, these metrics provide a comprehensive view of usability by combining measures of user performance with feedback about user satisfaction. According to Tullis and Albert (2023), evaluating both objective and subjective measures allows organizations to gain a more complete understanding of the user experience and make better-informed design decisions.

Supporting Evidence-Based Decisions

The selected testing methods and metrics support evidence-based decision making because they provide objective information about user performance and experience. For example, if users consistently require more time than expected to complete transfers, designers can investigate whether navigation or page layouts need improvement. If satisfaction ratings are low, additional research can identify features causing frustration. Moderated testing helps uncover the reasons behind usability problems, while unmoderated testing confirms whether those problems affect a broader user population. Together, these approaches provide reliable evidence that can guide design improvements and support stakeholder decisions.

To conclude, usability testing is an essential part of creating effective digital products. For the online banking application, moderated usability testing and unmoderated remote usability testing each offer valuable insights into the money transfer workflow. Moderated testing provides detailed feedback and a deeper understanding of user behavior, while unmoderated testing offers efficiency and larger sample sizes. By selecting appropriate metrics such as task completion rates, time on task, error rates, and user satisfaction scores, organizations can make evidence-based decisions that improve usability and enhance the overall user experience.

No AI-generated tools were used.

References

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Tullis, T., & Albert, W. (2023). *Measuring the user experience: Collecting, analyzing, and presenting usability metrics* (4th ed.). Morgan Kaufmann.